📘 REQUIREMENTS DOCUMENT

Product: LerenGO – Campus Network (MVP)

Version: 1.0

Timeline: 3 Weeks

Scope: Single University Launch

1️⃣ PRODUCT OVERVIEW

Campus Network is a university-specific digital platform that provides:

- Verified academic notes (admin-uploaded)
- Paid note unlocking via Razorpay
- Structured doubt posting
- Verified senior response system
- University-based access isolation

This release focuses only on one university.

2️⃣ USER ROLES

2.1 User

- Login via Google (Apple optional)
- Select university
- View notes for their university
- Unlock paid notes
- Post doubts
- View their payment history

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2.2 Senior (Verified)

- Has user permissions
- Can be assigned doubts
- Can respond to doubts
- Manually verified by admin

2.3 Admin

- Upload notes
- Assign note price
- Verify seniors
- Assign doubts
- View payments
- Manage universities

3️⃣ FUNCTIONAL REQUIREMENTS

3.1 Authentication

Requirements:

- Users must login via Google OAuth
- Email must be stored
- If first login → prompt university selection
- University_id must be saved in users table

Rules:

- User cannot proceed without selecting university
- Role defaults to “user”

3.2 University Isolation

Requirement:

Users must only access content belonging to their university.

Enforcement:

- All notes queries must filter by university_id
- All doubts must be linked to university_id
- Access control must be validated server-side

Failure = critical defect.

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3.3 Notes Management

Admin Requirements:

- Upload PDF
- Enter:
  - title
  - description
  - price (in paisa)
  - university_id
  - is_free
  - is_verified

User Requirements:

- View list of notes
- Free notes: download
- Paid notes: show unlock button

3.4 Paid Note Unlocking

Flow:

1. User clicks Unlock
2. Backend creates Razorpay order
3. Payment checkout opens
4. Razorpay webhook verifies
5. Payment marked success
6. note_access record created
7. Download enabled

Critical Requirement:

Unlock only after webhook verification.

Frontend success response must NOT unlock content.

3.5 Doubt System (MVP Lite)

User:

- Post doubt (text)
- View assigned senior
- View response

Admin:

- Assign doubt to verified senior

Optional:

- Paid doubt unlock (if time permits)

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3.6 Payment System

Requirements:

- Razorpay test mode for staging
- Razorpay live mode for production
- Webhook endpoint must:
    - Verify signature
    - Validate order_id
    - Update payment status
    - Create note_access record

Data stored:

- razorpay_order_id
- razorpay_payment_id
- amount
- status
- user_id
- item_type
- item_id

3.7 Storage Requirements

- Notes stored in private bucket
- Downloads via signed URL
- No public bucket access

3.8 Admin Panel

Admin must be able to:

- View all users
- Verify seniors
- Upload notes
- View payments
- Assign doubts

Admin routes must be protected by role check.

4️⃣ NON-FUNCTIONAL REQUIREMENTS

4.1 Security

- Webhook signature verification mandatory
- No secret keys in frontend
- Use environment variables
- Enforce university filtering server-side

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4.2 Performance

- Notes list load under 2 seconds
- Payment verification under 5 seconds
- Signed URL expiry ≤ 5 minutes

4.3 Reliability

- Payment must not unlock twice
- Duplicate webhook must not duplicate access
- Failed payment must not create note_access

4.4 Compliance

- Privacy policy page
- Terms of service
- Refund policy
- Support email visible

5️⃣ DATA MODEL (High-Level)

Entities:

User → belongs to → University

Note → belongs to → University

NoteAccess → belongs to → User + Note

Doubt → belongs to → University

Payment → belongs to → User

6️⃣ ACCEPTANCE CRITERIA

Launch ready only if:

- Users cannot access other university notes
- Paid notes unlock only after webhook
- Test payment works 10/10 times
- Signed URLs expire properly
- Admin panel secured
- No payment unlock via frontend manipulation

7️⃣ RISKS

- Payment fraud
- Webhook misconfiguration
- University isolation bug
- Public storage exposure

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Mitigation:

- Strict backend validation
- Repeated testing
- Manual QA

8️⃣ RELEASE READINESS CHECK

Before production:

- Razorpay live keys set
- Webhook URL updated
- Domain connected
- SSL active
- Environment variables verified
- Test real ₹10 payment successful